

Customer Segmentation Project

Customer Analytics using K-Means Clustering

Academic Project Report

Tools: Python, pandas, scikit-learn, HTML/CSS/JavaScript

1. Project Overview

This project segments customers using behavioral and demographic information. The goal is to identify groups with similar purchasing patterns so that marketing, retention, cross-selling and re-engagement strategies can be tailored to each group.

2. Objectives

The project aims to prepare customer data for analysis, apply K-Means clustering, evaluate alternative cluster counts, profile the resulting customer segments, visualize their characteristics and translate the findings into practical business recommendations.

3. Dataset

An illustrative dataset of 600 customers was created for academic demonstration. It contains age, gender, city, annual income, purchase frequency, average order value, annual spend, recency, online purchase ratio, discount usage, preferred category and loyalty score.

4. Methodology

Eight numerical features were standardized using StandardScaler. K-Means models with 3 to 6 clusters were compared using silhouette score. The selected solution used 3 clusters because it achieved the strongest silhouette score in the tested range. Cluster profiles were then translated into descriptive segment labels.

5. Segment Summary

Segment	Customers	Avg Spend (INR)	Frequency	Recency	Loyalty
High-Value Loyalists	129	64,643	25.0	34.4	82.5
At-Risk / Inactive	234	27,011	9.1	39.8	44.8
Deal-Driven Shoppers	237	18,915	11.8	34.4	54.8

6. Targeted Business Insights

High-Value Loyalists: protect retention with VIP rewards, premium launches and referral benefits.

Frequent Core Customers: increase basket size through bundles, cross-selling and personalized recommendations.

Deal-Driven Shoppers: use targeted promotions with minimum-order thresholds to reduce margin erosion.

Occasional Customers: use category-based recommendations and lifecycle reminders to increase purchase frequency.

At-Risk / Inactive: run win-back campaigns based on previous preferences and recency.

7. Dashboard

The accompanying interactive dashboard allows users to filter by segment, city and preferred category. It displays customer count, average annual spend, average purchase frequency, average loyalty score, segment distribution, spend by segment and a segment profile table.

8. Conclusion

Customer segmentation converts raw customer records into actionable groups. The project demonstrates data preparation, unsupervised machine learning, model evaluation, customer profiling, visualization and business insight generation. The resulting segments can support more relevant targeting and more efficient allocation of marketing effort.

Dataset note

All customer records in this submission are illustrative and generated solely for academic demonstration; they are not claimed to represent a real company or real individuals.